

# ControlPlane.ai — Round 2 Final

## Admission Control for AI That Acts

Accenture Innovation Challenge 2026 · Round 2 · dense hybrid (R2S1–R2S4)

**ETERNAL:** Default = `UNSUPPORTED` · entitlement = set-membership (zero LLM) · exact  $R \times S$  matrix · one graph · hard gate on actions · FNR as typed format · two-pending-actions (R1 Edit + R3 Escalate **held**, never “blocked”) · UNKNOWN never  $\rightarrow$  SUPPORTED · bias *async-only* · *refuse-to-claim* (about us)

Sources: docs/ARCHITECTURE.md · docs/reference/NARRATIVE.md · docs/reference/QA.md · docs/ps.md · Stages 1–4 absorbed in this FINAL (locks deleted after merge).

| Official deliverable (docs/ps.md) | Artifact                                            |
|-----------------------------------|-----------------------------------------------------|
| Detailed Business Proposal        | This document                                       |
| Working Prototype                 | controlplane/ · examples/ · tests/                  |
| Pitch Presentation                | R2S5.md + submission/ControlPlane_Round2_Pitch.pptx |

## Stage Check (compact)

| Stage                     | Locked into | Status |
|---------------------------|-------------|--------|
| R2S1 prototype scope      | §§1–4       | PASS   |
| R2S2 enterprise envelope  | §5          | PASS   |
| R2S3 build/demo           | §6          | PASS   |
| R2S4 buyers/value/rollout | §§7–11      | PASS   |

  

| # | Eternal invariant                                                                                                               | Status |
|---|---------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | Default = <code>UNSUPPORTED</code>                                                                                              | PASS   |
| 2 | Entitlement = <code>span.acl</code> $\subseteq$ <code>principal.clearance</code> ; zero LLM                                     | PASS   |
| 3 | Exact $R \times S$ matrix; never redrawn; no route parameter                                                                    | PASS   |
| 4 | One graph: <code>STEP</code> $\rightarrow$ <code>SPAN</code> $\rightarrow$ <code>CLAIM</code> $\rightarrow$ <code>ACTION</code> | PASS   |
| 5 | Hard gate on <b>actions</b> , not tokens                                                                                        | PASS   |
| 6 | Dual-action: R1 <b>Edit</b> + R3 <b>Escalate (held)</b> , never “blocked”                                                       | PASS   |
| 7 | UNKNOWN never $\rightarrow$ SUPPORTED                                                                                           | PASS   |
| 8 | FNR as typed format; empty until earned                                                                                         | PASS   |

| #  | Eternal invariant                                      | Status      |
|----|--------------------------------------------------------|-------------|
| 9  | Bias = async route-level only (never live matrix cell) | <b>PASS</b> |
| 10 | Refuse-to-claim list (about <i>us</i> )                | <b>PASS</b> |

## 1. Problem

Enterprises moved from AI that **answers** to AI that **acts**. Failure changed category: bad paragraph → **executed transaction**.

*“Approved. Refund of INR 1,84,000 issued under clause 7.2 of the vendor agreement.”*

**Clause 7.2 does not exist.** Failure is *absence* of evidence, not conflict. Filters pass it. Confidence can read 0.94. Money moved Tuesday, found Friday. If ungated, **the company wrongly pays out INR 1,84,000** — the customer did not lose money.

This is not “hallucination” as the market frames it. It is an **authorisation** problem: an unproven claim authorized an irreversible action. Solution is not better text scoring — it is **admission control**.

*The system didn’t fail. It was never asked to prove anything.*

*Everyone watches the exit. Nobody records the entrance.*

Three structural reasons existing approaches fail:

1. **Inspect output, not context contract.** Without recording what the model was *given*, verification is unfalsifiable opinion. With it, verification is set-membership: *which span proves this claim?*
2. **Score the response, not the action.** Groundedness 0.82 means the same on a draft and a wire transfer. One threshold cannot price both.
3. **Identity-blind.** The common enterprise incident is a **correct answer to the wrong person**. No output-only inspector carries caller identity into verification.

| Approach                                       | Why it fails here                                                               |
|------------------------------------------------|---------------------------------------------------------------------------------|
| Post-hoc observability (LangSmith, Arize, ...) | Traces <i>after</i> commit — audit trail, not interlock                         |
| LLM-as-judge / wrappers                        | Opinion, identity-blind, too slow for commit, cannot state own error rate       |
| Static guardrails                              | Miss lexically clean fabrications and ACL failures                              |
| RAG groundedness                               | Averages (one wrong figure drowns); action-blind; <b>retrieval ≠ permission</b> |
| Confidence / logprob                           | Named failure is <i>confidently</i> wrong                                       |
| Composite risk scores                          | Three owners/costs collapsed into one number that maps to no actuator           |

Failure modes this plane addresses: (1) categorical claim with **no provenance** authorizing R3; (2) claim bound to evidence the **caller may not read**; (3) **two pending actions, different blast radii** on one response; (4) no real-time ground truth → **invert burden of proof**.

| Overlap (brief)                         | Treatment on one graph                                           |
|-----------------------------------------|------------------------------------------------------------------|
| Fabricated personal detail, no span     | Unsupported + no-span PII/secret shape; stays <b>UNSUPPORTED</b> |
| Correct HR fact, ACL excludes caller    | Entitlement violation — independent of semantic correctness      |
| Absent policy clause authorizing refund | Unsupported categorical on R3 → Escalate ( <b>held</b> )         |
| Bias on decision-shaped outputs         | <b>Not</b> a per-response verdict; async flip-rate + CI          |

Multi-turn = more STEPs on the **same session ledger** — not a separate product. Prior assistant text is never evidence by reappearance. Unproven turn-1 claim used to authorize R3 in turn N remains unproven. Tool results enter as SPANs, not automatic authorization. Compounding risk = same unproven claim reaching higher R — already covered by the frozen graph.

Enterprise question is not “how do we score AI responses?” It is: *What evidence is required before a claim may authorize a specific action, for this caller, on this route?*

Buyer question a sceptic can answer without our slide: *What consequential actions does this route perform today, what is the loss if one is wrong, and what fraction can sit behind an earned admission boundary?*

## 2. Solution

ControlPlane.ai is an **admission-control layer** (firewall / transaction validator class — not observability, not a second model). Deployed as thin context-assembly SDK hook + OpenAI-compatible reverse proxy. No weights, logits, or fine-tuning. Integration cost is real — **and is the moat**.

AI output → claim proof → entitlement → R×S → action admitted / edited / escalated / held

One primitive, three reads:

STEP ■■■produces■■■ SPAN ■■■binds■■■ CLAIM ■■■authorizes■■■ ACTION

Performance reads forward. Cost reads backward (exact dead compute). Responsibility reads labels. One structure — not three detectors.

### Load-bearing primitives

1. **Provenance outside the model** at context assembly — keystone: `source_id · ACL · content_hash · offsets` + calling principal. Model has **no write path**. Binding computed by the plane; model-emitted citations are not evidence; no open-web rescue.
2. **Default = UNSUPPORTED**. Earn SUPPORTED via recomputation or binding. Unsupported ≠ low confidence — **unproven**. UNKNOWN never → SUPPORTED.
3. **Claim-type routing**. Numeric/date/ID → recompute. Textual → bind (entailment, not string match). Derived/multi-hop → recompute or UNKNOWN.
4. **Entitlement = set-membership** (CALLER → CLAIM → SPAN → SOURCE ACL; `span.acl ⊆ principal.clearance`). Zero LLM. Lane 1 always on. Semantically correct + unauthorized still fails. **Most differentiated mechanism** — competitors cannot replicate without carrying identity into verification.
5. **Exact R×S matrix per pending action**. Disposition = **worst claim for that action — never an average**. R = irreversibility × audience × data class × autonomy. Action Interlock = sole final decider; pure

rule engine.

6. **Hard gate on actions, not tokens.** Text hold-back ~150–300 ms. Speculative verification OK; **speculative release forbidden.**

7. **Escalate** → **evidence packet** (claim, spans, verdict, diff). Surgical Edit only (strip or one constrained regen; re-gate; second fail → Escalate).

8. **Publish own per-route FNR as typed format** — empty until earned. Emptiness is the credibility play.

**Ungrounded / parametric:** when no provenance spans exist for a route, it is declared ungrounded by construction; such claims **cannot authorize any action** regardless of blast radius. *We don't claim to verify what we were never given. We claim what we were never given cannot authorize an action.*

### Refuse-to-claim (about us, not competitors)

1. **“We eliminate hallucinations.”** False if shipped. Narrower claim: ungrounded claims cannot authorise actions, and we report what we miss.

2. **“Zero integration, drop it in.”** We hook context assembly — real work, exact reason the design works. Scope = one SDK hook + OpenAI-compatible proxy (days on a standard retrieval stack, not quarters) — never sold as drop-in. Integration cost is the moat.

3. **“Zero added latency.”** We never make the model feel slow; we make the action wait. Quote only **≤40 ms p50 / ≤200 ms p95** — never 40 as p95.

4. **“One accuracy number across failure modes.”** Hallucination, leakage, and bias have different mathematics, error costs, and owners.

**Operating claim:** ControlPlane does not promise “safe” or “truthful” AI. It makes an unproven or unauthorized claim unable to authorize an action beyond the route’s admitted boundary — and publishes what the plane missed.

## 3. Matrix

Exact frozen 16-cell R×S — transcribed, never redrawn, **no route parameter:**

|    | Contradicted / entitlement violation | Unsupported categorical | + | Unsupported hedged | + | Unknown         |
|----|--------------------------------------|-------------------------|---|--------------------|---|-----------------|
| R3 | Block                                | Escalate                |   | Escalate           |   | Escalate        |
| R2 | Block                                | Edit                    |   | Edit               |   | Escalate        |
| R1 | Edit                                 | Edit                    |   | Pass + annotate    |   | Pass + annotate |
| R0 | Pass + annotate                      | Pass + annotate         |   | Pass               |   | Pass            |

**R tiers:** R0 = internal draft · R1 = user-visible read-only · R2 = reversible write / external send · R3 = irreversible or regulated (payment, deletion, publication, regulated advice).

**Actuators (exact):** Block · Edit · Escalate · Pass / Pass + annotate. Forbidden inventions: STREAM, Kill Span, Hold & Re-verify, Redact & Flag, COMMIT BLOCKED as R3 unsupported-categorical label. Autonomy downgrade / circuit breaker = spoken controls, not inventable demo labels.

**Fail stance ≠ matrix:** Matrix answers “evidence arrived and is bad → actuator?” Fail stance answers “evidence did not arrive in time?” Floors: R0/R1 fail **open + annotate**; R2/R3 fail **closed or escalate**.

Universal fail-open forbidden.

#### Centrepiece cells (simultaneous on one refund response):

| Pending action | Tier      | Finding                              | Cell                         | Actuator                           |
|----------------|-----------|--------------------------------------|------------------------------|------------------------------------|
| Show text      | <b>R1</b> | Unentitled span grounds a claim (C3) | R1 × entitlement             | <b>Edit</b>                        |
| Issue refund   | <b>R3</b> | Clause 7.2 has no span (C2)          | R3 × unsupported-categorical | <b>Escalate — held with packet</b> |

Never collapse into one response-level verdict. **Never say “blocked” about the refund.** Same unsupported+categorical claim yields R1→Edit on a draft and R3→Escalate on a payment — proof scales with consequence; the matrix is not renamed severity.

## 4. Prototype

Exactly **two** live routes. Third live decision-support / bias route **refused**. Bias = proposal measurement only. Scope tension (resolved): brief names decision-support and bias; architecture defines bias as distributional/async — both stay in the **enterprise envelope**, never contaminate the live core.

| Route                         | Dominant R          | Mechanism proved              | Live?                     |
|-------------------------------|---------------------|-------------------------------|---------------------------|
| <b>A. Refund agent</b>        | R1 text + R3 refund | Two-pending-actions           | <b>YES</b>                |
| <b>B. Knowledge assistant</b> | R0/R1               | Provenance + entitlement flip | <b>YES</b>                |
| Decision-support / bias       | —                   | Async counterfactual          | <b>NO — envelope only</b> |

**Refund risk signature:** absent clause · numeric/ID error · entitlement leak into customer text · irreversible payment on same response · no real-time ground truth. Latency: R0/R1 text **≤40 ms p50 / ≤200 ms p95** (never quote 40 as p95); action verify amortised in tool RTT; text hold-back ~150–300 ms. Speculative verification OK; speculative release forbidden.

**Knowledge:** correct-but-unauthorized disclosure; fabrication; ACL-excluded span. Demo stages ACL-excluded → R1 × entitlement = Edit (not no-span PII → Block). Plane enforces carried ACLs — does **not** repair IAM. Authorization path: CALLER → CLAIM → SPAN → SOURCE ACL — set-membership, **zero LLM**.

**Why this pair:** Refund proves one graph, two actuators, hard gate, surgical edit, packet. Knowledge proves graph ≠ RAG groundedness: same claim, different principal → different outcome (RAG averages; ControlPlane prices worst claim per pending action). Together: *is this claim proven?* and *is this caller entitled to that proof?* A third live bias route adds async measurement without strengthening the dual-action centrepiece.

**Assumptions (load-bearing):** synthetic enterprise-shaped corpora only — no real PII; every span carries source/ACL/hash/offsets; generator via OpenAI-compatible API only (no weights/logits); decision policy = pure rule engine; traffic directional tens of thousands/week (prototype = curated single-node — no throughput claim); FNR null without trustworthy ground truth; corpus poisoning outside truth guarantee (hash makes source auditable, not factually true); DPDPA/GDPR as config packs — not certification.

## Boundary — in

| Prove                           | How                                                           |
|---------------------------------|---------------------------------------------------------------|
| Provenance outside model        | Spans before claims; model cannot author                      |
| One typed ledger                | STEP → SPAN → CLAIM → ACTION                                  |
| Default UNSUPPORTED             | Earn via bind/recompute; UNKNOWN never → SUPPORTED            |
| Claim-type routing              | Numeric recompute; textual entailment; derived → UNKNOWN      |
| Deterministic entitlement       | Set-membership; zero LLM                                      |
| Exact 16-cell matrix            | Per pending action; worst-claim weighting                     |
| Dual-action centrepiece         | R1 Edit + R3 Escalate held                                    |
| Hard action gate                | Mock refund committed: false while Escalate                   |
| Surgical Edit + evidence packet | Strip/regen once; packet = claim+spans+verdict+diff           |
| Empty FNR schema                | Typed nulls only — no fabricated %                            |
| Principal flip                  | Change only caller → outcome flips                            |
| Per-claim surface               | Verified / Uncertain / Blocked; refund action = Held/Escalate |
| Ungrounded / parametric         | No provenance → cannot authorize any action                   |
| Live compute                    | Binding/entitlement/interlock visible (~20–80 ms)             |

## Boundary — out

Third live bias route · per-response bias · production load as mechanism proof · real payments/PII · live IAM repair · LLM-as-judge primary · confidence/risk scores · open-web truth · generative rewrite · fabricated FNR · Lane-3 on critical path · triage SLA UI · weights/logits · collapsing dual-action · calling refund “blocked.”

## Measured results (this build)

All numbers are produced by `make eval` and `make bench` and regenerate on every run — they are measured, never asserted. The capability ledger and per-route Wilson CIs live in the generated README PDF (`submission/ControlPlane_Round2_README.pdf`) and `evals/last_run.json`.

**This build earns the FNR on a 168-case self-authored corpus.** The committed run (`evals/last_run.json`) reports:

**Ungrounded FNR 1.1%** (95% Wilson CI 0.2%–5.8%; n=94 ungrounded claims; 1 published miss).

We publish the interval — the honest claim is the interval, not a fake-perfect 0% and not a tighter bound than the data supports.

**Passable-action FPR 0.0%** (95% Wilson upper bound 15.5%; n=21 passable claims).

**Hard-negative hold rate 64% [50.7%, 75.7%].** 53/168 cases are hard negatives (31.5%, above the 20% floor). We over-flag — that is our named next milestone. Clean strata hold 0/13.

**One published miss:** `struct-miss-000` — a response citing “Clause 4.1 permits this refund”

where the span only says clause 4.1 *covers shipping delays*. Structural symbol lookup matches → SUPPORTED → low-tier action slips through. We show the judge exactly which case we got wrong.

**bullet No production proof.** Production FNR is unknown until shadow replay over live traffic (ARCHITECTURE §7, §12). The honest claim: "on this self-authored corpus we miss 1.1% of ungrounded claims (95% Wilson CI 0.2–5.8%; n=94); production is unknown."

**bullet Refuse-to-claim:** we do not eliminate hallucinations; we do not claim drop-in integration; we do not claim zero added latency; we do not claim one accuracy number; we do not fill FNR with fabricated percentages.

Shape of the published claim (ARCHITECTURE §7): we hold **X%** of ungrounded responses at **Y ms** p50 — and here is the **Z%** we don't, with a Wilson interval.

## 5. Enterprise Envelope

Same plane on every route. Runtime:

```
App/Agent → Context SDK Hook (SPAN: source·ACL·hash·offsets)
           → OpenAI-compatible proxy
           → Evidence Ledger STEP→SPAN→CLAIM→ACTION
           → Action Interlock exact R×S = f(R,S) [no route param]
           → Pass / Pass+annotate / Edit / Escalate / Block
           → Text hold-back · Action commit boundary
```

**Shared (not soft-configurable):** Provenance Recorder · Ledger · typed claims · proof paths · verdict set · Entitlement Auditor · Interlock · exact matrix · fail-stance floors · hard gate · surgical edit · packet · claim-level surface · FNR schema · Lane 1 always on.

**RoutePolicy one-liner (configures budget, not truth):**

```
RoutePolicy { route_id · tenant · use_case · provenance_scope · action_grammar
  · action_to_R_mapping (locked R3: payment|deletion|publication|regulated_advice)
  · verification_profile · fail_stance_by_R (may NOT loosen R2/R3 to fail-open)
  · enforcement_mode shadow|canary|enforce · error_budget · escalation_target
  · sampling_policy · geography_overlay (additive only) · latency_budget ≤40/≤200 }
```

Routes may change actions, R-mapping (except locked classes), Lane-2 budget, shadow/enforce, sampling, additive geo. Routes may **not** change: UNSUPPORTED default · entitlement · matrix cells · hard gate location · UNKNOWN→SUPPORTED · composite scores · Stage 1 dual-action scope. Low-consequence traffic gets **less verification budget and a proportionate actuator — not weaker truth semantics**.

| Route profile           | Dominant R         | Lane posture                             | Enforcement                                                                    |
|-------------------------|--------------------|------------------------------------------|--------------------------------------------------------------------------------|
| Customer-support refund | R1+R3 simultaneous | Lane 1 text; Lane 1+2 + interlock for R3 | Shadow/canary before enforce; <b>demo still shows hard gate on mock refund</b> |
| Internal knowledge      | R0/R1              | Lane 1 dominant; ACL always inline       | Enforce after shadow evidence                                                  |



| Route profile    | Dominant R                | Lane posture                      | Enforcement                                                          |
|------------------|---------------------------|-----------------------------------|----------------------------------------------------------------------|
| Decision-support | R0/R1 memo + R2/R3 action | Lane 2 derived; Lane 3 async bias | Action gate when earned; bias always async — <b>not Stage 1 live</b> |

| Lane     | Runs                                                           | Path                              |
|----------|----------------------------------------------------------------|-----------------------------------|
| <b>1</b> | Span membership, ACL, arithmetic, typed interlocks, PII shapes | Inline; always on; no LLM         |
| <b>2</b> | NLI/binding for flagged textual / high-R                       | Near-line; timeout → UNKNOWN      |
| <b>3</b> | Semantic-entropy, bias replay, calibration, shadow audit       | Async; <b>never</b> critical path |

**Governance:** versioned DAG of (signal, threshold, action, latency\_budget); decision time = pure rule engine. Hierarchy narrows only: Enterprise baseline → geo overlay (additive) → tenant → route → action class. Overlays cannot loosen frozen invariants. Global parse-time invariants include: default UNSUPPORTED · UNKNOWN→SUPPORTED forbidden · entitlement always on · Lane 1 cannot disable · locked R3 classes · matrix immutable · R2/R3 fail closed/escalate · hard gate on actions · no composite score · no LLM at decision · bias never a matrix verdict · Stage 1 scope unreopened.

**Policy lifecycle:** Draft (hashed) → static validation → shadow replay → canary → auto-rollback if override >3× baseline or error budget breached → named-principal approval → promote. Day-one = **shadow**. Production enforce is **earned**.

**Audit:** principal · evidence · claim verdict · matrix cell · actuator · policy\_version · verifier\_versions · latency · lane · route\_id — reconstructible from ledger + policy + frozen matrix.

**Feedback = calibration, not adaptation.** Learn thresholds/error-budget candidates via shadow→canary. Never learn: confidence/logprobs · silence-as-safety · aggregate override as permission to loosen · poisoned sources · prior assistant text as evidence · LLM-as-judge opinions · online weight updates. Override = evaluation evidence, not real-time security rewrite.

**FNR schema (null until earned):**

```
route_id · policy_version · evaluation_window · strata_definitions
· sampled_count_per_stratum · false_negative_count · ground_truth_positive_count
· FNR_estimate · CI_lower · CI_upper · ground_truth_method
· measurement_status # null|insufficient_sample|prototype_corpus|production_measured|stale
· limitations
```

Method: stratified shadow audit — 100% of Block/Escalate/Edit + sample of Pass; ground truth = human / expensive multi-verifier — **never** LLM-as-judge. Claim shape when earned: “On this route we catch \<measured\>% of ungrounded claims at 40 ms p50 — and here is the \<measured\>% we don’t.”

**Bias (brief requirement, frozen stance):** async route-level counterfactual flip-rate + CI; flag when CI excludes zero. **Never** claim → bias verdict → matrix. Never dropped; never a Stage 1 live route.

| Layer  | Enterprise                               | Stage 1 live       | Proposal-only |
|--------|------------------------------------------|--------------------|---------------|
| Routes | Support + knowledge + decision + agentic | <b>Exactly two</b> | Extra packs   |



| Layer             | Enterprise                           | Stage 1 live                            | Proposal-only   |
|-------------------|--------------------------------------|-----------------------------------------|-----------------|
| Core              | Full plane                           | Dual-action + flip + ledger + empty FNR | Multi-tenant HA |
| Bias / FNR values | Async + measured when earned         | Absent / null placeholders              | Full programs   |
| Scale             | Tens of thousands/week (directional) | Curated single-node traces              | Load/HA         |

**Prototype = proof the plane works. Envelope = proof the plane can be operated. Pitch = both.**  
Envelope residual risks (architecture): (1) derived/multi-hop false SUPPORTED → recompute-or-UNKNOWN + stratified FNR; (2) poisoned sources / wrong ACLs → hash + entitlement-by-source detector — defend claim↔evidence link, not source truth; (3) over/under-flag + R mis-mapping → shadow earn-out, locked R3, interlock in executor. Stage 2 IS configuration/governance/lifecycle around the primitive. Stage 2 IS NOT a second mechanism, redrawn matrix, live bias product, or trained safety model.

## 6. Build & Demo

### Corpora (essentials — no breadth)

| Source            | ACL                                        | Purpose                                         |
|-------------------|--------------------------------------------|-------------------------------------------------|
| AGR-VENDOR-v3     | agent-readable                             | Clauses 1–6 only; <b>no clause 7.2 anywhere</b> |
| ORD-1023          | refund_agent                               | amount=184000 INR — C1 SUPPORTED                |
| FIN-INTERNAL-NOTE | internal_analyst (excludes agent_refund_7) | C3 entitlement → Edit                           |
| INJECT-NOTICE     | untrusted                                  | Cannot author provenance                        |
| HR-COMP-L6        | hr_partner                                 | Principal-flip span                             |

Principals: agent\_refund\_7 · analyst\_01 · hr\_partner\_01. Canonical fixture: “*Approved. Refund of INR 1,84,000 issued under clause 7.2 of the vendor agreement.*” + one FIN-INTERNAL sentence. Span contract: span\_id · source\_id · ACL · content\_hash · offsets · text · step\_id.

| Claim                                        | Finding                                                             | Feeds                               |
|----------------------------------------------|---------------------------------------------------------------------|-------------------------------------|
| <b>C1</b> amount/order numeric               | Binds ORD-1023 → <b>SUPPORTED</b>                                   | Proof works                         |
| <b>C2</b> “under clause 7.2 ...” categorical | <b>No span</b> → stays <b>UNSUPPORTED</b> (absence ≠ contradiction) | refund.execute → R3 <b>Escalate</b> |
| <b>C3</b> grounded on FIN-INTERNAL           | ACL excludes caller → <b>entitlement</b>                            | text.show → R1 <b>Edit</b>          |

All three born UNSUPPORTED. Optional polish only after dual-action+flip green: paraphrase entailment · parametric ungrounded · numeric near-miss — cut polish before cutting dual-action.

## Components

Real: Recorder · Ledger · Claim Extractor (fixture) · Numeric Recomputer · Entitlement Auditor · Interlock · Surgical Editor · Packet · Hold-back · Policy loader · Principal switch · FNR renderer · Trace Console · harness. Thin mock: generator stub · refund executor (REFUND HELD / committed:false — never COMMIT BLOCKED) · NLI labels. No LLM at decision time. No confidence field in the decision path.

## Demo order (≤8 min; UI ≥60% ledger)

**Governing test:** if removing the graph leaves the demo looking the same, scope failed. Open on held transaction — never a risk statement. Core crisis ≤90s. First spoken sentence about failure contains **claim**, not response.

| Must show                                                       | Must NOT show                                                                   |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| Ledger ≥60%; spans before claims; matrix cells before actuators | Composite risk/confidence scores                                                |
| Action Gate cold-open with committed boolean                    | "Response blocked" / COMMIT BLOCKED for R3 unsupported-categorical              |
| Per-claim Verified/Uncertain/Blocked; refund = Held/Escalate    | LLM-as-judge pane · open-web lookup                                             |
| Evidence packet on every Escalate; empty FNR                    | Interlock bypass override · third-route chrome · bias widget · chatbot-majority |

1. **Cold open gate:** refund.execute {amount:184000, reason:"clause 7.2", order\_id:"ORD-1023"} · R3 · **HELD — ESCALATE** · executed:false
2. Expand ledger — spans with source·ACL·hash **before** claim verdicts
3. C1 SUPPORTED · C2 no span UNSUPPORTED · C3 entitlement violation
4. Highlight cells **before** actuators: text.show → R1×entitlement→**Edit**; refund.execute → R3×unsupported-categorical→**Escalate**
5. Surgical Edit strips C3; refund stays held; packet opens for C2 (claim, candidate spans [], verdict, diff)
6. Executor proves not committed — company does **not** wrongly pay out in the gated demo
7. Empty FNR schema visible (null placeholders only)
8. **Principal flip (≤2 min):** analyst\_01 → Edit; flip only to hr\_partner\_01 → Pass. Zero LLM.

Optional third beat (cut first): same unsupported claim as R1 vs R3 → actuator changes solely because R changed — or numeric 14 vs span 11 → R1 Edit.

Voice: authorise · admit · prove · bind · refuse · hold · escalate · gate — not monitor · detect · observe · watch · guard · trust score · risk score.

## Run

```
python3 -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"
python3 -m pytest tests/ -v
python3 examples/refund_trace_demo.py
python3 examples/knowledge_flip_demo.py
```

| Command                             | Proves                                                                                       |
|-------------------------------------|----------------------------------------------------------------------------------------------|
| <code>pytest tests/ -v</code>       | Matrix, entitlement, UNSUPPORTED default, hard gate, held≠blocked, R2S1 criteria on fixtures |
| <code>refund_trace_demo.py</code>   | Dual-action: Edit + Escalate held; never collapsed; never “blocked”                          |
| <code>knowledge_flip_demo.py</code> | Same span/claim; outcome flips with clearance only                                           |

## 7. Buyers

### Target users

Beachhead = high-consequence routes (refund-class R3 + mixed-governance knowledge) — not “all enterprise AI safety.”

| Role                       | Who                                       | Buys / operates                                                                                                                                               |
|----------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economic buyer</b>      | Ops / CS / CHRO / CRO / CFO / CISO / Risk | Action authorisation + per-route error budgets. Responds to: <i>unproven claim cannot authorise an action</i> . Not: <i>we detect 95% of hallucinations</i> . |
| <b>Technical buyer</b>     | Platform / ML infra / Identity            | SDK hook + proxy + adapters; wants graph, not second black box; ≤40/≤200 ms                                                                                   |
| <b>App / agent team</b>    | Route owners                              | Wire adapters; no app rewrite                                                                                                                                 |
| <b>Day-to-day actor</b>    | Support / knowledge worker                | Sees Verified/Uncertain/Blocked per claim; Held/Escalate on action — never enforces                                                                           |
| <b>Operator / governor</b> | Route owners, SRE, Risk Ops               | Shadow/canary/enforce; rollback; circuit breaker; ledger reconstruct                                                                                          |
| <b>Escalation reviewer</b> | Risk / ops                                | Evidence packet, not bare alert                                                                                                                               |
| <b>Compliance / audit</b>  | Legal / DPO / audit                       | Influences via reconstructible trail                                                                                                                          |

Split that matters: who **pays when it fails** ≠ who **runs it** ≠ who **types the answer**. Interlock enforces; plane publishes miss rate.

## 8. Impact Logic

### Business case

No fabricated ROI. No “99%.” No net-savings slide. Value = **mechanism** → **consequence**; buyer fills middle terms. Tail risk, not average case.

$$\text{Exposure} = \text{freq}(\text{consequential AI actions}) \times P(\text{unproven/unauthorized claim}) \times \text{loss/wrong action}$$

| Lever                                          | Mechanism → consequence                                                                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A — Avoided wrong actions</b> (primary)     | R3 Escalate held (executed:false) is structural escape. Value = held true-positives × buyer's direct cost of that action class; residual sized by published FNR. Artifact: attempted commit · matrix cell · gap · committed=false. |
| <b>B — Blast-radius pricing</b>                | Lane 1 majority volume; expensive proof where R justifies. Same verdict annotates a draft and holds a payment. Answers brief's one-size-fits-all latency problem.                                                                  |
| <b>C — Exact dead compute</b>                  | Backward walk: STEP grounding zero accepted claims = waste (exact). Observability measures spend; this measures waste. Expose number; enterprise prices own traffic — no % saved on slide.                                         |
| <b>D — Alert fatigue without lowering gate</b> | Matrix prices actuators; R0/R1 unsupported+hedged → Pass+annotate. Shadow earn-out. Metrics: overrides · gate-fail · edit/escalation <b>per route</b> .                                                                            |
| <b>E — Auditability</b>                        | Hash-chained ledger + versioned policy. Reconstruct: action → cell → verdict → span → source/hash/ACL → principal → policy → latency. Regulator answer = pointer.                                                                  |
| <b>E2 — Unauthorized disclosure</b>            | ACL-excluded span caught when semantically correct. Does not fix IAM — stops IAM gaps being <b>silently bypassed by a model</b> .                                                                                                  |
| <b>F — Publish misses</b>                      | Per-route FNR typed format; null until earned. Conversation: route · population · miss · CI — not “trust our safety score.”                                                                                                        |
| <b>G — Earned autonomy</b> (secondary)         | Shadow → enforce only where counterfactuals justify. More AI action <b>after</b> evidence — never lead claim.                                                                                                                      |

## 9. Roadmap

Earn-out, not feature calendar. Enforcement earned, not switched on. Day-one = shadow. Misses published every phase.

| Phase                | What                                                       | Exit                                            |
|----------------------|------------------------------------------------------------|-------------------------------------------------|
| <b>0 — Prototype</b> | Two live routes; synthetic corpora; ledger ≥60%; empty FNR | All R2S1/R2S3 criteria; dual-action crisis ≤90s |

| Phase                            | What                                                                                                                                    | Exit                                                          |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>1 — Shadow production</b>     | 1–2 real routes matching pattern; dual-emit; no production hold yet. Determinism works day one; statistics earn thresholds              | Counterfactuals enough to open canary                         |
| <b>2 — Canary R0/R1</b>          | Full policy lifecycle; enforce Edit/Pass+annotate on read-only                                                                          | Tolerable ops; no material edit regression                    |
| <b>3 — Limited R2/R3 enforce</b> | Hard-gate selected actions; packets to humans; fill FNR only with trustworthy GT; locked R3 at parse                                    | Buyer FNR/FP/override thresholds; committed=false on held TPs |
| <b>4 — Broader envelope</b>      | More routes (decision-support <b>template</b> ); additive geo; async bias; dead-compute FinOps; circuit breaker; fail stance tier-owned | Multi-route without second detector                           |

## 10. Risks

| Risk                                       | Why it matters                                                                 | Mitigation                                                                                                    |
|--------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| False assurance on derived/multi-hop       | Shallow entailment → false SUPPORTED — worse than no plane                     | Bypass NLI → recompute or UNKNOWN; UNKNOWN never SUPPORTED; timeout→matrix; FNR by claim type                 |
| Poisoned sources / wrong ACLs              | Plane proves claim↔evidence + carried ACLs — not source truth / not IAM repair | Immutable source+hash; missing ACL→unentitled; entitlement-violation <b>by source</b> ; quarantine via policy |
| Prompt injection / model-declared bindings | Invent evidence edges                                                          | Plane computes bindings; model cannot author spans; disposition = rule engine                                 |
| Over-flag → bypass; under-flag → liability | Classic guardrail death                                                        | Shadow default; blast-radius pricing; fail-stance floors; override>3× rollback; circuit breaker               |
| R mis-mapping (payment as R1)              | Wrong row without “breaking” matrix                                            | Locked R3 at parse; interlock in <b>executor</b> , not only UI                                                |
| Integration / provenance gaps              | Incomplete spans → false UNSUPPORTED                                           | Honest hook+proxy; evidence coverage measured; conservative high-R                                            |
| Plane as SPOF                              | Outage / load-induced bypass                                                   | Tier-owned fail stance; universal fail-open forbidden                                                         |

| Risk                                  | Why it matters                           | Mitigation                                                                                      |
|---------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------|
| Feedback misuse / weight access       | Train on overrides; require logits       | API-only; feedback→policy candidates only; calibration ≠ trained judge                          |
| Switch-off after a quarter            | Layers that interrupt low-blast text die | Gate on actions; majority Pass+annotate; earn-out before trust; integration cost stated as moat |
| Pattern-match as RAG/safety dashboard | Differentiation dismissed                | Open on held INR 1,84,000; ledger majority UI; ban monitor/detect/risk-score vocabulary         |

## 11. Differentiation

| ControlPlane                                 | Everyone else                                     |
|----------------------------------------------|---------------------------------------------------|
| Provenance outside model at context assembly | Inspect output; treat model citations as evidence |
| Default = UNSUPPORTED                        | Default allow; flag what looks wrong              |
| Entitlement = ACL set-membership             | Identity-blind scorers                            |
| One graph, three reads                       | Three bolted tools                                |
| Exact R×S per pending action                 | Composite risk / confidence                       |
| Hard gate on actions                         | Gate on text/tokens                               |
| Publish per-route FNR (misses)               | Publish precision (bother rate)                   |

### Ordered contrast

1. **vs observability** — post-hoc traces after harm. Observation without commit control is audit trail, not architecture. ControlPlane records entrance and interlocks commit; same graph yields exact dead compute.

2. **vs LLM-as-judge / static guardrails** — *does this look right? vs which span proves it?* Decision time = pure rule engine. Entitlement = identity, not classification. Default UNSUPPORTED is posture, not threshold tweak.

3. **vs RAG groundedness** — closest cousin, still short: retrieval-only, averages, action-blind. ControlPlane binds full provenance, prices **worst claim per action**, same unsupported claim → R1 Edit vs R3 Escalate. **Retrieval ≠ permission. Proof scales with consequence.**

**Closer:** Competitors publish precision. ControlPlane publishes misses and **refuses** to claim it eliminates hallucination/bias/privacy. Almost none disclaim themselves. Plane audited by the standard it enforces.

## 12. Fidelity + Content Laws

| Invariant                                                       | Status    |
|-----------------------------------------------------------------|-----------|
| Default = UNSUPPORTED                                           | Untouched |
| Entitlement / ACL set-membership; zero LLM                      | Untouched |
| Exact R×S; no route parameter                                   | Untouched |
| Hard gate on actions; speculative release forbidden             | Untouched |
| Dual-action R1 Edit + R3 Escalate <b>held</b> (never “blocked”) | Untouched |
| FNR typed format; empty until earned                            | Untouched |
| UNKNOWN never → SUPPORTED                                       | Untouched |
| No LLM-as-judge / confidence as primary path                    | Untouched |
| Bias = async route-level only                                   | Untouched |
| Refuse-to-claim (about <i>us</i> )                              | Untouched |
| Exactly two Stage 1 live routes                                 | Untouched |
| Latency ≤40 ms p50 / ≤200 ms p95                                | Untouched |
| Surgical edit · evidence packet · Lane 1 always on · locked R3  | Untouched |

| Law                    | Exact rule                                                                                                                          |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Clause 7.2</b>      | Does <b>not</b> exist. Absence ≠ conflict. Never “caps/denies/doesn’t cover.” → Unsupported+category → <b>Escalate</b> , not Block. |
| <b>Held ≠ blocked</b>  | Refund <b>held and escalated with the evidence packet</b> — never “blocked.”                                                        |
| <b>Who pays</b>        | <b>Company wrongly pays out.</b> Customer did not lose money.                                                                       |
| <b>Dual action</b>     | Text → R1×entitlement→Edit (C3). Refund → R3×unsupported-category→Escalate (C2). Both simultaneous.                                 |
| <b>Latency</b>         | ≤40 ms p50 / ≤200 ms p95. Never quote 40 as p95. Speculative verify OK; speculative release forbidden. Hold-back ~150–300 ms.       |
| <b>Refuse-to-claim</b> | No: eliminate hallucinations · zero integration · zero added latency · one accuracy number.                                         |

**Closing spine:** AI can act → unproven claim must not authorize → provenance outside model → Default UNSUPPORTED → entitlement set-membership → R×S prices by consequence → hard gate on commit → publish misses.

Vocabulary: authorise · admit · prove · bind · refuse · hold · escalate · gate — not monitor · detect · observe · watch · guard · trust score · risk score · “responsible AI” as virtue.

*Once you accept that an AI response is a set of claims requesting permission to act, you must capture provenance outside the model, invert the burden of proof, carry identity into verification, and gate the*



*commit path. Any softer design is a different product.*

## Appendix — Official brief coverage

| Brief requirement (docs/ps.md)                           | Where                     | Freeze stance                                              |
|----------------------------------------------------------|---------------------------|------------------------------------------------------------|
| Different risk/latency by use case                       | §5 RoutePolicy + profiles | Same graph/matrix; config only                             |
| Overlapping bias/hallucination/privacy                   | §§1, 5                    | Not one classifier                                         |
| No reliable real-time ground truth                       | §§1–2; FNR null           | Honesty over bluff                                         |
| Over-/under-flag tradeoff                                | §§5, 8–10                 | Shadow earn-out; blast-radius pricing                      |
| Multi-turn / agents that act                             | §§1, 5 session ledger     | No separate product                                        |
| Evolving geo/industry regulation                         | §5 additive overlays      | Cannot loosen matrix                                       |
| API-only foundation models                               | §§2, 6                    | No weights/logits                                          |
| Detection options (AI-as-judge, confidence)              | §§2, 11                   | <b>Reject</b> as primary                                   |
| Decision logic allow/edit/flag/block                     | §3 matrix                 | Transcribed                                                |
| Architecture placement                                   | §§2, 5–6                  | Pre-commit interlock + hold-back                           |
| Governance / audit                                       | §5                        | Zero LLM at decision time                                  |
| Feedback loops                                           | §5                        | Calibration, not trained model                             |
| Metrics / FP/FN / trustworthiness                        | §§5, 8F                   | Publish misses as format                                   |
| Multi use-case, tens of thousands/week, mixed governance | §§4–5                     | Directional; prototype simulated                           |
| Deliverables: proposal + prototype + pitch               | Header map                | Proposal=this ·<br>Prototype=controlplane/ ·<br>Pitch=next |

*End of ControlPlane.ai Round 2 Dense Final Hybrid — Stage Check PASS. Pitch from this file + live demo.*